From LSTM to Transformer: Comparing and Improving Architectures for Clinical Accuracy in Chest X-Ray Report Generation

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Abstract

Medical imaging is a high-growth area for image captioning, where models aim to "read" X-rays and suggest preliminary findings. In this project, we explore how different architectures, from hierarchical LSTMs to Transformer-based models impact both linguistic quality and clinical correctness, and investigate methods to reduce hallucinated findings through clinical-term guidance. We compare three decoders on IU X-Ray under a shared training pipeline and evaluate both linguistic overlap (BLEU, METEOR, ROUGE-L, CIDEr) and clinical content (CheXbert F1). We find that clinical guidance shifts the model along a fluency–specificity trade-off, improving CheXbert F1 and CIDEr at modest cost to BLEU, with softer guidance settings outperforming stronger ones. We also find that joint encoder–decoder training is essential at this dataset scale.

1. Introduction & Background

When a radiologist reads a chest X-ray, they must translate visual patterns into precise written language and that report drives clinical decisions. In busy hospitals, a single radiologist may read hundreds of scans per day [5], making it a tedious and repetitive task. We built and compared three automated systems that can read a chest X-ray: a hierarchical LSTM [5], a standard Transformer [10], and a clinically guided Transformer augmented with mechanisms designed to encourage medically meaningful vocabulary. Our central question is whether linguistic fluency and clinical accuracy can be achieved simultaneously, or whether there is an inherent trade-off between them.

The current dominant approach follows an encoder-decoder pattern: a convolutional network extracts image features, and a text decoder generates the report word by word [2, 5]. The field has shifted from recurrent decoders toward Transformers augmented with memory [2], region-level detection [9], and clinical terminology alignment [8]. At the frontier, large pretrained language models achieve strong results but require significant compute and cloud access which is incompatible with privacy-constrained clinical settings.

Two main limits persist across approaches. First, standard text metrics (BLEU, ROUGE-L) measure word overlap but cannot distinguish "no pneumothorax" from "pneumothorax" which are diagnostically opposite statements. Clinical labelers like CheXbert [7] address this, but many models still optimize only for fluency. Second, models tend to learn a fluent normal-finding template, producing generic reports that score well linguistically but miss or hallucinate specific findings [8]. Our work will implement and compare three models from scratch, specifically targeting a practical constraint: can a compact model, without large pretrained language models, achieve acceptable performance on both fluency and clinical correctness, or whether there is an inherent trade-off between them? If yes, it opens a realistic path toward local, privacy-preserving deployment.

2. Data

We train and evaluate all three models on the Indiana University Chest X-Ray (IU X-Ray) corpus, using the preprocessed split from Chen et al.'s R2Gen [1] which is generally used by recent radiology report generation papers (published baselines).

Composition: Each instance pairs one chest X-ray image with one free-text radiology report. Most studies contain two views (frontal/PA and lateral) stored as separate PNG files sharing the same report. The dataset comprises 2,955 studies corresponding to 5,910 images.

Splits: We use the canonical R2Gen train/val/test partition (2069/296/590 studies) (Table 4). Split assignments are stored in `annotations.json` and used by all three dataloaders to ensure no study leaks between splits.

Preprocessing:

Images: resized to 224×224 and normalized with ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]), compatibility with the pretrained DenseNet-121 encoder.

Text: reports are lowercased and whitespace-normalized. Tokenization replaces each period with a `<SEP>` boundary token before splitting on whitespace; commas, exclamation marks, and question marks are stripped. `<SEP>` serves a dual purpose: hierarchical models use it to identify sentence boundaries, while non-hierarchical models treat it as an ordinary token. Since Vocabulary.decode() strips `<SEP>` before scoring, all three models produce BLEU-visible output equivalent to standard period based tokenization.

Vocabulary: built from the training split only. Words appearing fewer than 3 times are mapped to `<UNK>`, yielding 975 tokens (including special tokens `<PAD>`, `<BOS>`, `<EOS>`, `<UNK>`, `<SEP>`).

Sequence format: reports are wrapped with `<BOS>`/`<EOS>` and truncated to 60 tokens. Teacher-forcing offset is applied at load time: `input_tokens` drops the final token, `target_tokens` drops the first, so `input[t]` predicts `target[t]`. Batches are right-padded to the longest sequence; padding is masked in all three losses via `ignore_index=pad_id`.

Limitations: With only 2,069 training studies, it is small by modern standards and thus models may underfit relative to their published performance, and seed-to-seed variance is higher. The corpus also reflects the case mix of a single US academic medical center, skewing toward normal studies. Rare findings are underrepresented, and models may learn to generate plausible-sounding normal reports regardless of image content.

3. Approach

3.1. Overall Experiment Design

Our approach centers on a controlled comparison of three decoder architectures: a hierarchical LSTM, a standard Transformer decoder, and a clinical-term-guided Transformer. To isolate the effect of the decoder, all three share the same visual encoder — a DenseNet-121 [6] pretrained on ImageNet — and are trained on identical data splits with the same optimizer, learning rate schedule, and evaluation pipeline. Differences in output quality can therefore be attributed to the decoder architecture and guidance mechanism rather than to feature extraction or training procedure. Our experimental contribution is the clinical-term-guided Transformer, which augments a standard Transformer decoder with two mechanisms for clinical correctness: weighted cross-entropy loss and bias-constrained decoding.

3.2. Model 1: Hierarchical LSTM

The Hierarchical LSTM serves as the recurrent baseline in our three-way comparison. Jing et al. [5] were among the first to treat report generation as a two-level structured task rather than a single-stream captioning problem: reasoning at the sentence level before committing to individual words mirrors how radiologists write, and the architecture has since become the standard recurrent baseline against which Transformer-based systems measure their gains [3, 11].

Our implementation follows Jing et al. [5] with three adaptations: (1) DenseNet-121 replaces the original VGG encoder to align with the CheXNet-style features used across all compared models; (2) dual-view encoding is added so both frontal and lateral views can be consumed from the standard R2Gen batch format; and (3) the dataloader interface is unified with the shared 5-tuple batch format.

A pretrained DenseNet-121 encodes each image into a spatial map of shape (B, 1024, 7, 7) for 224×224 inputs. A global context vector $\mathbf{v} \in \mathbb{R}^{512}$ is produced via adaptive average pooling, linear projection, and batch normalization. When both views are present, per-view spatial maps and global vectors are mean-pooled across the view dimension; the spatial map then feeds the co-attention module, and the global vector feeds the sentence LSTM.

At each sentence step s, soft spatial attention [11] computes a context vector over the spatial feature map conditioned on the current sentence LSTM hidden state $\mathbf{h}_s$:

$$\alpha_i = \text{softmax}\left(\mathbf{W}_a \tanh\left(\mathbf{W}_v \mathbf{f}_i + \mathbf{W}_h \mathbf{h}_s\right)\right), \mathbf{c}_s = \sum_i \alpha_i \mathbf{f}_i$$

where $\mathbf{f}_i \in \mathbb{R}^{1024}$ is the i-th spatial location.

A single-layer sentence LSTMCell unrolls for $s_{\text{max}} = 6$ steps, taking $[\mathbf{c}_s; \mathbf{v}]$ as input and producing a topic vector $\mathbf{t}_s$ plus a binary stop prediction. A two-layer word LSTM then generates one sentence per topic vector, with $\mathbf{t}_s$ initializing both hidden and cell states via learned projections.

The training loss is $\mathcal{L} = \mathcal{L}_{\textbf{stop}} + \mathcal{L}_{\textbf{word}}$, computed only over active (non-padding) sentence slots with `ignore_index=pad_id`. At inference, a sentence slot is skipped if the stop head predicts class 1; the word LSTM decodes greedily per sentence, and per-sentence outputs are concatenated for evaluation by `pycocoevalcap` and CheXBert.

3.3. Model 2: Transformer

The vanilla Transformer serves as the architectural base for Model 3's clinical guidance (Section 3.4).

Visual encoder bridge: The DenseNet-121 backbone produces a feature map of shape (B,1024,7,7). We flatten this to a sequence of 49 visual tokens and project to `d_model` via a learned linear layer, yielding the key/value memory for the decoder's cross-attention. Visual tokens receive no positional embedding; the decoder recovers spatial structure through cross-attention alone. The encoder is trained end-to-end with the decoder in all reported runs.

Decoder and training: We use PyTorch's `nn.TransformerDecoderLayer` configured with `d_model`=512 , `n_heads`=8, `d_ff`=2048, `dropout`=0.3, and either 3 or 6 layers (evaluated in Section 4.3). Token embeddings are summed with sinusoidal positional encodings [10]; input and output embeddings are not tied. Training uses teacher forcing with unweighted cross-entropy and `ignore_index=PAD_ID`; a causal mask prevents attention to future positions. At inference we decode greedily up to a 60-token cap, freezing each sample once it emits `<EOS>` until all samples in the batch terminate.

3.4. Model 3: Clinically Guided Transformer

The clinical-term-guided Transformer [2] extends the vanilla standard Transformer baseline [10] with two mechanisms designed to make the decoder lean toward clinically accurate language. The base architecture is identical to model 2, ensuring that any performance difference is attributable to the guidance mechanisms alone.

Clinical vocabulary construction: We curated a clinical lexicon by mapping each of the 14 CheXpert finding categories [4, 7] to tokens present in our 975-word vocabulary. For each category, we identified seed words capturing the core finding and its common synonyms as they appear in radiology reports. For example, "Pleural Effusion" maps to tokens including "pleural," "effusion," while "Cardiomegaly" maps to "cardiomegaly, "heart". Beyond the finding terms, we identified two additional tiers: negation tokens and hedge tokens. In total, approximately 130 unique tokens out of 975 received clinical annotations. Table B shows complete examples of this mapping across finding, negation, and hedge tiers.

Mechanism 1 - Weighted cross-entropy loss: We construct a weight vector of dimension equal to the vocabulary size, where each position holds a multiplier: finding tokens receive weight α_{finding}, negation tokens receive α_{negation}, and all other tokens retain a weight of 1.0. During training, this vector is passed to CrossEntropyLoss as the `weight` parameter. The model then will incur a higher gradient signal when it mistakenly predicts a clinical term (high penalty), encouraging it to learn the distribution of medical vocabulary more precisely. This aims to address the hallucination problem in prior work [8]. We treat the weight multipliers as hyperparameters and explore several configurations: *hard* (α_{finding}=3.0, α_{negation}=2.0), *soft* (α_{finding}=2.0, α_{negation}=1.5), *no weighting* (α_{finding}=1.0, α_{negation}=1.0) as the baseline.

Mechanism 2 - Constrained decoding with logit bias: During inference, before the softmax at each decoding step, we add a positive bias to the logits corresponding to clinical term positions. The biased logits are computed as $\mathbf{l}' = \mathbf{l} + \beta \cdot \mathbf{m}$ where m is a binary mask over the vocabulary (1 for clinical terms, 0 otherwise) and β is a configurable bias strength. This shifts the probability mass toward clinical tokens without changing the trained model weights, making the model more likely to produce medically relevant words. This approach is inspired by approaches that incorporate retrieved clinical terminology during generation [8]. We explore bias strengths of β=0.0 (off), β=0.5 (moderate), and β=1.0 (strong). Negation tokens can optionally be included in or excluded from the bias mask.

3.4.1 Ablation design

We yield a 2×2 ablation matrix with four configurations that toggle each on or off to isolate the two mechanism's contribution. Beyond this core ablation, we also test whether increasing decoder depth from 3 to 6 layers interacts with the guidance mechanisms, hypothesizing that a higher-capacity decoder may better leverage clinical term signals without collapsing to short generic outputs.

3.5. Evaluation Methodology

We evaluate all models using two families of metrics: **linguistic metrics** that assess text quality and **clinical metrics** that assess diagnostic correctness.

Linguistic Metrics: We report BLEU-1 through BLEU-4, METEOR, CIDEr, and ROUGE-L, computed using the standard `pycocoevalcap` package. These metrics assess text quality from complementary angles — n-gram precision, synonym-aware recall, informativeness weighting, and subsequence overlap, respectively. Detailed definitions are provided in Appendix C. Linguistic metrics are computed on the validation set after every epoch during training and on the held-out test set for final evaluation.

Clinical Metrics: Linguistic metrics can score a report highly even when it is clinically incorrect — for example, "no pneumothorax" and "pneumothorax is present" share most words but are diagnostically opposite. To address this, we evaluate clinical correctness using CheXbert [7], a BERT-based labeler that maps each report to a 14-dimensional label vector corresponding to the CheXpert finding categories [4], with each finding classified as positive, negative, uncertain, or unmentioned. We run CheXbert on both generated and ground-truth reports and compute precision, recall, and F1 per finding, macro-averaged F1, hallucination rate (false positive findings), and missed finding rate (false negatives).

3.6. Anticipated challenges

We identified some risks before implementation. First, IU X-Ray has only 4,138 training images, small by deep-learning standards; we expected the Transformer to be particularly prone to overfitting and planned to mitigate with dropout, weight decay, early stopping, and the option to freeze the pretrained encoder. Secod we expected linguistic and clinical metrics could move in opposite directions, motivating the dual-evaluation design. Third, our clinical vocabulary risked limited coverage — medical terms appearing in radiology reports may not survive the minimum-frequency filter (`min_freq`=3), mapping important clinical terms to the unknown token.

4. Experiments and Results

4.1. Implementation and training details

All models share a common training pipeline. All models were implemented in PyTorch using nn.TransformerDecoderLayer for the Transformer decoder and nn.LSTMCell for the hierarchical LSTM. Each decoder receives the same DenseNet-121 spatial features described in Section [3]. We trained with AdamW (lr $1e{-}4$, weight decay 0.01), 2-epoch linear warmup followed by cosine decay to $0.1\times$ base lr, gradient clipping at norm 1.0, and mixed-precision (fp16). Batch size was 16; padding tokens were excluded from the loss via `ignore_index`. Transformer runs were budgeted at 30 epochs and the LSTM at 50, with early stopping on validation loss (patience 10) applied throughout; most runs terminated before the budget. All test-time decoding was greedy. All runs used `seed=42`; compute constraints precluded multi-seed averaging, limitation described in Section 4.6.

4.2. Initial experiments and what didn't work

We summarize the key dead-ends and corrections per architecture after tuning.

Hierarchical LSTM: smaller is better, but only with the right schedule. The default configuration from [5] (`hidden_size`=512, `word_num_layers`=2, `s_max`=6, `dropout`=0.1) reached validation BLEU-1 of 0.215. A higher-capacity variant (`hidden_size`=768, `word_num_layers`=3, `s_max`=8, `dropout`=0.3) collapsed to BLEU-1 of 0.078 — with only 4,138 training images, the larger LSTM overfit aggressively despite heavier dropout. Returning to the smaller architecture, stronger regularization (dropout 0.2) and a lower learning rate ($5e{-}5$) both *hurt* validation performance (BLEU-1 0.171 and 0.143). The model had moved from overfitting to underfitting.

The configuration that worked combined moderate dropout (0.15), longer warmup (3 epochs), and a longer training budget (50 epochs) at the original learning rate. This reached validation BLEU-1 of 0.286 — a 33% improvement over the baseline — and a lower training loss ($\sim$0.73 vs. $>$1.1). On a dataset of this size, the bottleneck was *schedule and training duration*, not capacity.

Vanilla Transformer: encoder unfreezing was decisive. Our initial configuration froze the DenseNet-121 encoder. Training loss decreased normally but validation BLEU-4 reached only 4×10^{-6}: the model produced short "everything normal" templates regardless of input (e.g., *"heart size and mediastinal contour are normal pulmonary vascularity is normal..."*) — a known failure mode where frozen ImageNet features lack the discriminative signal to distinguish abnormal from normal chest X-rays.

Unfreezing the encoder produced the largest single improvement in our study: validation BLEU-4 jumped to $\sim$0.13 and test BLEU-4 to 0.103. Reports lengthened (30–50 tokens vs. 10–15) and included specific findings (*"stable left basilar atelectasis versus scarring"*). We take this unfrozen 6-layer Transformer (dropout 0.1) as our reference vanilla configuration.

Clinical-guided Transformer: more guidance is not better. We hypothesized that explicitly guiding the model toward clinical vocabulary would improve CheXbert F1. The empirical picture proved more nuanced: guidance contributes real but partial gains, and stronger settings consistently underperform softer ones (Section 4.4). However, *high-strength guidance produced mode collapse*. Our first clinical Transformer (3 layers, dropout 0.3, $\alpha_{\text{finding}} = 3.0$, $\beta = 1.0$) reached training loss of 0.92 but produced 10–15-token reports of short generic phrases; validation BLEU-4 was effectively zero. Outputs were grammatically correct but too short for 4-gram matches. $\alpha = 3.0$ reshapes the loss landscape such that emitting a few high-confidence clinical tokens followed by an early end-of-sequence is locally optimal, and $\beta = 1.0$ compounds this at inference — a degenerate but loss-minimizing strategy.

Softening guidance recovered fluency. We reduced decoder dropout to 0.1 (matching the vanilla baseline) and softened both strengths to $\alpha = 2.0$, $\beta = 0.5$. At 6 layers (`soft_6Layer`), test BLEU-4 reached 0.084 and CIDEr 0.507 — the highest CIDEr in our study. Reports lengthened to 25–35 tokens. Guidance also produced visible signatures: specific findings absent from no-guidance variants (*"degenerative changes in the thoracic spine"* on CXR1004) and negation enumerations attributable to the decoding-bias mechanism. More details are discussed in section 4.4

4.3. Main results

We compare the headline configuration of each architecture: hierarchical LSTM lstm_hierarchical_optimized (Section 4.2); vanilla Transformer (6-layer, encoder unfrozen, Section 4.2); and clinical-guided Transformer clinical_soft_6layer ($\alpha_{\text{finding}} = 2.0$, $\beta = 0.5$, Section 4.2). Table 1 reports linguistic-overlap and clinical-content metrics together.

LSTM trails both Transformers decisively with lowest BLEU-4 and magnitude lower CIDEr. We attribute this not to under-tuning — LSTM_run_optimized was the result of an extended schedule and capacity sweep, but to the fun-

Model	Linguistic-overlap						Clinical-content		
	B-1	B-2	B-3	B-4	MET	R-L	CIDEr	$\mathrm{ChX}_{\mathrm{macro}}$	$\mathrm{ChX}_{\mathrm{micro}}$
lstm_hierarchical_optimized	0.183	0.094	0.056	0.034	0.085	0.169	0.040	0.138	0.831
vanilla_Transformer_6L_unfrozen	**0.378**	**0.230**	**0.152**	**0.103**	**0.161**	**0.310**	0.354	0.189	**0.923**
clinical_Transformer_soft_6layer	0.269	0.164	0.114	0.084	0.135	0.279	**0.507**	**0.233**	**0.923**

Table 1. Test-set results on IU X-Ray. Best per-column value bolded. **B-**n = BLEU-n, **MET** = METEOR, **R-L** = ROUGE-L, **ChX** = CheXbert F1. CheXbert micro-F1 is dominated by *no_finding* cases and saturates for both Transformers; macro-F1 is the more informative clinical metric.

damental difficulty of producing long, specific reports with a hierarchical recurrent decoder on 4,138 training images.

The two Transformers sit at different points on a fluency–specificity trade-off. The vanilla Transformer wins every linguistic-overlap metric, beating the clinical Transformer by roughly 0.02–0.04 on each BLEU-n. The clinical Transformer wins both clinical-content metrics: CIDEr 0.507 vs. 0.354 (+43%) and CheXbert macro-F1 0.233 vs. 0.189 (+23%). The two models tie on CheXbert micro-F1.

This was not what we initially expected. The vanilla Transformer's BLEU-4 of 0.103 is broadly consistent with comparable single-decoder models reported on this dataset, our most effective single intervention was jointly training the encoder rather than freezing it. The clinical Transformer trades fluency for specificity: it produces shorter, less reference-template-aligned reports that more often include specific finding terms and negation enumerations, which CIDEr and CheXbert reward. A loss-weighting-only ablation variant achieves slightly higher BLEU than our headline configuration; Section 4.4 dissects this trade-off mechanism by mechanism.

4.4. Ablation: dissecting the guidance mechanisms

The clinical-guided Transformer combines two mechanisms whose individual contributions are not isolated from the headline numbers in Section 4.3. We ran an 11-configuration ablation across two architectures (3 and 6 decoder layers, both at dropout 0.1, encoder unfrozen) and two guidance strengths (*hard*: $\alpha_{\text{finding}} = 3.0$, $\alpha_{\text{negation}} = 2.0$, $\beta = 1.0$; *soft*: $\alpha_{\text{finding}} = 2.0$, $\alpha_{\text{negation}} = 1.5$, $\beta = 0.5$). At each architecture we evaluated the four cells of the 2×2 ablation (loss weighting on/off $\times$ decoding bias on/off), with one additional cell per architecture for soft/hard comparison. Table 2 reports the 3-layer results; the corresponding 6-layer table appears in Appendix D and the central trade-off is visualized in Figure 1.

Hard guidance is uniformly worse than soft guidance. At 3 layers, hard guidance (D) reaches BLEU-4 = 0.073 and ChX-$\mathrm{F1}_{\mathrm{macro}}$ = 0.240; soft guidance ($\mathrm{D}_{\mathrm{soft}}$) reaches 0.076 and **0.267**, improving on *both* axes simultaneously. At 6 layers (Appendix D), hard wins clinical-F1 by 0.012 but loses BLEU-4 by 0.011 — a wash. We attribute this to the failure mode in Section 4.2: high-strength

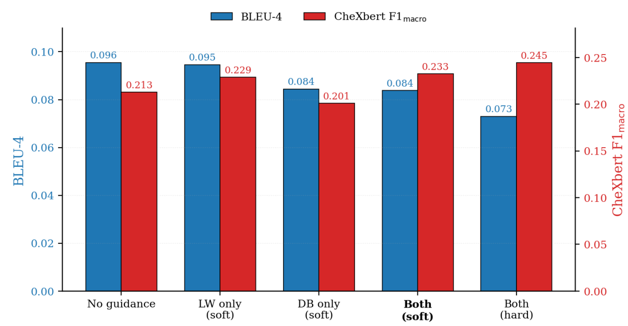

Figure 1. Linguistic vs. clinical metric trade-off

guidance reshapes the loss landscape such that emitting a few high-confidence clinical tokens followed by an early EOS becomes locally optimal, while soft guidance preserves enough standard language-modeling signal for fluent generation.

Doubling decoder depth does not uniformly help. 6 layers beats 3 layers on BLEU-1 in every matched cell, but the gap shrinks or reverses on macro-F1 — the best 3-layer cell ($\mathrm{D}_{\mathrm{soft}}$, 0.267) outperforms every 6-layer cell except 6L hard guidance (0.245). With only 4,138 training images, the deeper decoder has more capacity to fit the dominant "everything normal" template that BLEU-4 rewards; the 3-layer decoder cannot fit it as cleanly and emits more variable output, which costs BLEU but better matches finding diversity.

Decoding bias alone is the worst single mechanism. The 3-layer DB-only cell (C) has the lowest BLEU-1 (0.209), BLEU-4 (0.066), CIDEr (0.328), and METEOR (0.114) — worse than no guidance on every metric except macro-F1. The same pattern holds at 6 layers (Figure 1). Decoding bias adds a constant logit term at *inference* only, but the model has been trained without that bias and produces a token distribution the bias pulls off-manifold. Loss weighting alone, by contrast, is the most cost-effective single mechanism: at 6 layers, `soft_finding_only` matches the no-guidance BLEU-4 within 0.001 (0.095 vs. 0.096) while improving CIDEr by 24% and macro-F1 by 7.5%.

Synthesis. Our headline configuration `soft_6Layer` is not the BLEU-optimal point — `soft_finding_only`

Cell	LW	DB	α_f	β	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr	macro-F1
A	off	on	–	–	0.263	**0.083**	**0.128**	0.267	0.346	0.193
B	on	off	3.0	–	0.246	0.082	0.127	**0.274**	**0.460**	0.232
C	off	on	–	1.0	0.209	0.066	0.114	0.257	0.328	0.232
D	on	off	3.0	1.0	**0.268**	0.073	**0.133**	0.268	0.410	0.240
D_{soft}	on	on	2.0	0.5	0.244	0.076	0.126	0.270	0.454	**0.267**

Table 2. 3-layer ablation. Cells: A = no constraints, B = LW only (hard), C = DB only (hard), D = both (hard), D_{soft} = both (soft). All runs share architecture (dropout 0.1, encoder unfrozen). *LW* = loss weighting; *DB* = decoding bias; $\alpha_f = \alpha_{\text{finding}}$; β = bias strength. Best per-column value bolded.

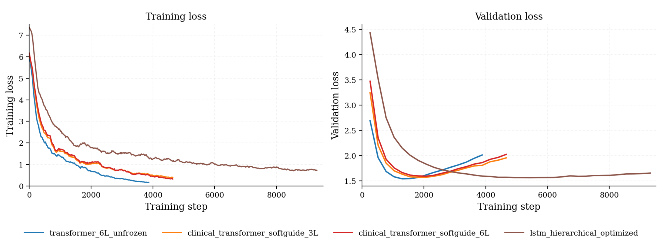

Figure 2. Training loss (left) and validation loss (right) for the LSTM (best run), vanilla Transformer (6L, encoder unfrozen), and the two soft-guidance clinical Transformers at 3 and 6 layers. full size in Appendix 3

is slightly higher — but it implements the full intended system, ties for best CIDEr (0.507), and beats the loss-weighting-only variant on macro-F1 (0.233 vs. 0.229). The practical recommendation: use soft loss weighting as the default, and add decoding bias only when clinical-finding F1 matters more than fluency.

4.5. Training dynamics

Figure 2 plots training and validation loss for the four headline runs. Two observations explain how training shaped our final results.

Transformers overfit; the LSTM plateaus. All three Transformer configurations bottom out on val_loss within 1500–2000 steps and rise steadily after. The vanilla Transformer reaches the lowest val_loss (1.54 at step 1300) but rises most steeply, while the two clinical Transformers bottom slightly later (1.58 at step 2000) and rise more gradually — consistent with the loss-weighting term acting as an implicit regularizer. The two clinical curves are nearly parallel: doubling decoder depth from 3 to 6 layers shifts val_loss only marginally, matching the test-set finding (Section 4.4) that depth's effect is small and inconsistent. The LSTM converges to 1.55 and plateaus rather than overfitting — for this architecture–dataset combination, capacity rather than overfitting risk is the bottleneck.

Encoder unfreezing was the single largest training-dynamics shift. Our initial frozen-encoder vanilla Transformer (not shown in Figure 2) reached training loss of 0.22 with validation BLEU-4 near 4×10^{-6} — training loss alone gave no warning. Unfreezing the encoder lengthened generated reports from 15 to 35 tokens within a few epochs and pushed validation BLEU-4 over 0.13. The lesson: training-loss curves are not sufficient diagnostics for sequence-generation models; per-epoch sample inspection was the intervention that made our development cycle tractable.

4.6. Conclusion and limitations

We compared three decoder architectures for chest X-ray report generation under a controlled training pipeline. Encoder unfreezing was our single largest improvement; the clinical-guided Transformer produced a measurable fluency–specificity trade-off, with softer guidance consistently outperforming stronger settings and loss weighting alone providing the most cost-effective improvement. The headline finding is that linguistic and clinical metrics do not move together: the vanilla Transformer wins BLEU/METEOR/ROUGE-L while the clinical Transformer wins CIDEr and CheXbert F1.

Several caveats apply. **Single seed:** all numbers come from `seed=42`; differences in the 6-layer ablation (BLEU-4 0.073–0.096; CheXbert F1_{macro} 0.201–0.245) fall within plausible seed variability, so we treat directional findings as more reliable than point estimates. **Hard/soft not fully crossed:** the soft-vs-hard comparison was only run on the both-mechanisms-on cell of each architecture, so the claim does not extend to the single-mechanism cells. **Greedy decoding:** we do not test beam search, which may behave differently with the decoding-bias mechanism. **Pipeline:** each view is processed independently (CXR1004 in Table 7 shows the resulting multi-view inconsistency); reported numbers come from re-evaluating the best-by-loss checkpoint with a uniform harness.

Future work could explore multi-seed evaluation and larger datasets such as MIMIC-CXR to assess whether the fluency–specificity trade-off persists at scale. Additionally, replacing greedy decoding with beam search and using CheXbert F1 as a reinforcement learning reward signal are directions to explore for improving both fluency and clinical correctness simultaneously.

Student Name	Contributed Aspects	Details
Yinghou Wang	Environment Setup, Implementation and Analysis	Designed and deployed the collaborative development environment (GitHub, PACE, W&B) and authored interface contract documentation. Built the shared model-agnostic training pipeline, configuration system, and checkpoint-resume infrastructure. Implemented the clinical-term-guided Transformer model and curated the clinical vocabulary from the CheXpert ontology. Trained and tuned both the clinical and vanilla Transformer models. Designed and conducted the clinical guidance ablation study.
Mohammad Yousuf Qureshi	Implementation and Analysis	Implemented the hierarchical LSTM baseline including the sentence-level and word-level decoder architecture. Trained and tuned the LSTM with hyperparameter exploration and analyze its performance. Implemented the linguistic evaluation module (BLEU, METEOR, CIDEr, ROUGE-L) and the clinical evaluation module (CheXbert F1, hallucination and missed finding rates).
Murad Magdiyev	Ideation, Data and Implementation	Originated the project concept and problem. Developed the data preprocessing pipeline including tokenization, vocabulary construction, and R2Gen canonical split integration. Implemented the vanilla Transformer decoder architecture.

Table 3. Contributions of team members.

5. Github Repository

The code for our project can be found here: https://github.com/vermouthwang/dl-cxr-report-gen https://gatech.box.com/s/pncft68i6os7hmw5om7aejr1n46ovamp

6. Work Division

Our work distribution can be seen in Table 3.

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A. Appendix

Split	**Studies**	**Images**
Train	2,069	4,138
Val	296	592
Test	590	1,180
Total	*2,955*	*5,910*

Table 4. R2Gen Data Split

B. Complete Clinical Lexicon

C. Evaluation Metric Definitions

Table 6 summarizes the evaluation metrics used in this work.

D. 6-layer ablation table

E. Headline Models Training and Validation Loss Curve

Table 5. Complete clinical lexicon used by the clinical-guided Transformer (Section 3.4). For each of the 14 CheXpert finding categories plus the negation and hedge tiers, we list the curated seed words. Italicized words marked † did not appear in the 975-token training vocabulary (failed the `min_freq=3` filter) and are excluded from the weight vector and bias mask at runtime. The lexicon contains 110 unique seed words; 99 (90%) are present in the vocabulary.

Category	**Seed words**
Findings (14 CheXpert categories)	
No Finding	normal, unremarkable, clear, negative, stable, intact, *patent*†, *preserved*†
Enlarged Cardiomediastinum	cardiomediastinal, mediastinal, mediastinum, enlarged, widened, widening, prominence, prominent
Cardiomegaly	cardiomegaly, heart, cardiac, enlarged, cardiopulmonary, cardiophrenic
Lung Opacity	opacity, opacities, opacification, density, densities, haziness, *shadow*†, *shadowing*†
Lung Lesion	lesion, lesions, mass, masses, nodule, nodules, nodular, fibronodular
Edema	edema, vascular, congestion, interstitial, vasculature, vasculatures, engorgement
Consolidation	consolidation, consolidations, consolidating, consolidative, infiltrate, infiltrates, airspace
Pneumonia	pneumonia, infection, infectious
Atelectasis	atelectasis, collapse, subsegmental, volume
Pneumothorax	pneumothorax, pneumothoraces
Pleural Effusion	pleural, effusion, effusions, fluid, pleura, subpleural, pleuroparenchymal
Pleural Other	thickening, scarring, scar, scars, fibrosis, fibrotic, calcification, calcifications, calcified, calcific
Fracture	fracture, fractures, rib, ribs, vertebral, *compression*†
Support Devices	catheter, tube, line, device, stent, valve, endotracheal, tracheostomy, picc, lines, clips, *clip*†, *wires*†, *pacemaker*†
Auxiliary tiers	
Negation	no, without, negative, normal, clear, unremarkable, stable, intact, *patent*†
Hedge	may, could, possible, possibly, suggests, suggestive, appears, consistent, concerning, questionable, borderline, *might*†, *likely*†

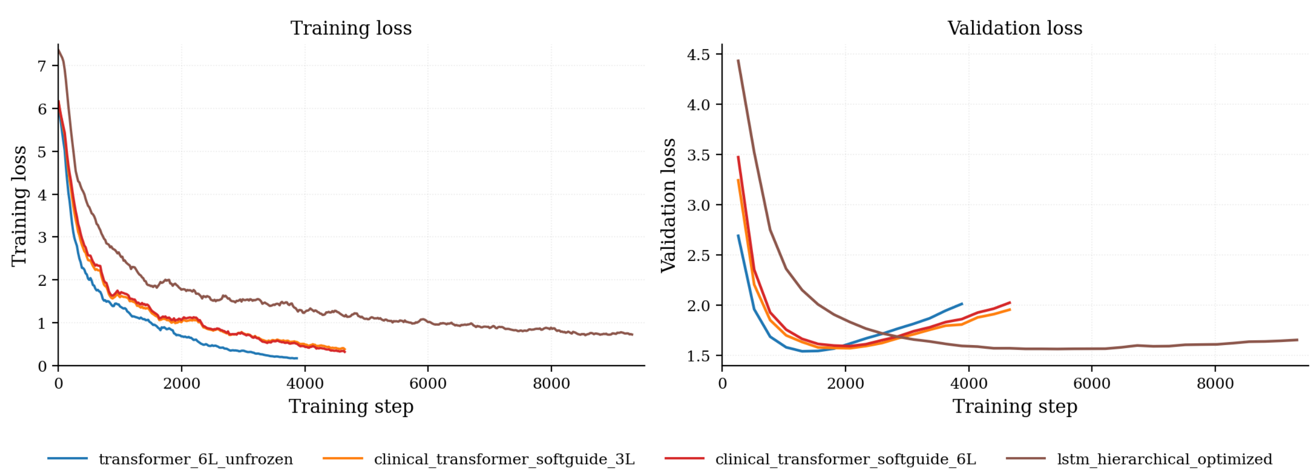

Figure 3. Training loss (left) and validation loss (right) for the LSTM (best run), vanilla Transformer (6L, encoder unfrozen), and the two soft-guidance clinical Transformers at 3 and 6 layers.

Family	Metric	Description	Range
Linguistic	BLEU-n (n=1..4)	Measures n-gram precision between generated and reference text. Higher n requires longer matching phrases; BLEU-4 effectively penalizes very short or repetitive outputs.	[0, 1]
Linguistic	METEOR	Computes alignment between generated and reference text using exact matches, stemming, and synonym matching via WordNet. Balances precision and recall with greater weight on recall.	[0, 1]
Linguistic	CIDEr	Uses TF-IDF weighting on n-grams so that rare, informative terms (e.g., "cardiomegaly") contribute more than frequent filler phrases (e.g., "the image shows"). Designed specifically for image captioning evaluation.	[0, 10]
Linguistic	ROUGE-L	Measures the longest common subsequence (LCS) between generated and reference text, capturing sentence-level fluency and word ordering without requiring contiguous n-gram matches.	[0, 1]
Clinical	CheXbert F1	Precision, recall, and F1 computed per finding over the 14 CheXpert categories, plus macro-averaged F1. Measures whether the generated report agrees with the reference on actual clinical findings, independent of wording.	[0, 1]
Clinical	Hallucination rate	Fraction of positive findings in the generated report that are absent from the ground truth. Measures how often the model invents diagnoses that do not exist in the reference.	[0, 1]
Clinical	Missed finding rate	Fraction of positive findings in the ground truth that are absent from the generated report. Measures how often the model fails to mention real diagnoses.	[0, 1]

Table 6. Evaluation metrics used for report generation assessment. Linguistic metrics measure text similarity between generated and reference reports; clinical metrics measure diagnostic correctness via structured label extraction.

Image ID	Reference	Hierarchical LSTM	Vanilla Transformer (6L)	Clinical Transformer (D_6L, soft)
CXR1004_IM-0005/0	the heart pulmonary xxxx and mediastinum are within normal limits there is no pleural effusion or pneumothorax there is no focal air space opacity to suggest a pneumonia the aorta is tortuous and ectatic there are degenerative changes of the acromioclavicular joints …	of the lungs are clear of the right hemidiaphragm is unchanged of the right hemidiaphragm is unchanged of the right hemidiaphragm is unchanged of the right hemidiaphragm is unchanged of the right hemidiaphragm is unchanged	the heart is normal in size and contour the lungs are clear without evidence of infiltrate there is no pneumothorax or effusion there is mild degenerative changes of the thoracic spine	the heart is normal in size the mediastinum is unremarkable the lungs are clear
CXR1004_IM-0005/1	*(same patient, second view)*	of the right chest xxxx with tip in the right atrium of the right chest xxxx with tip at the cavoatrial junction of the right chest xxxx with tip at the cavoatrial junction of the right chest wall xxxx xxxx represents a ¡UNK¿ of the right hemidiaphragm is again noted of the right hemidiaphragm is again noted	the lungs and pleural spaces show no acute abnormality heart size and pulmonary vascularity within normal limits	heart size and mediastinal contour normal pulmonary vascularity normal lungs are clear no pleural effusions or pneumothoraces degenerative changes in the thoracic spine
CXR1015_IM-0001/1	streaky and patchy bibasilar opacities <UNK> density projected over the heart on the lateral view no definite pleural effusion seen no typical findings of pulmonary edema … stable cardiomediastinal silhouette with normal heart size	of the lungs are clear of the right chest xxxx with tip at the cavoatrial junction of the right chest wall xxxx xxxx represents a ¡UNK¿ of the right chest wall with tip overlying the right atrium of the thoracic spine appears intact of the thoracic spine appears intact	the heart is normal in size and contour the lungs are clear without evidence of infiltrate there is no pneumothorax or effusion there is no acute bony abnormalities	heart size normal lungs are clear xxxx are normal no pneumonia effusions edema pneumothorax adenopathy nodules or masses
CXR1030$_{I}M - 0024/0.png$	streaky and patchy bibasilar opacities <UNK> density projected over the heart on the lateral view no definite pleural effusion seen no typical findings of pulmonary edema … stable cardiomediastinal silhouette with normal heart size	of the cardiac silhouette xxxx xxxx are normal of the right hemidiaphragm normal in contour of the right clavicle of the thoracic spine of the thoracic spinet	the heart size and cardiomediastinal silhouette are normal there is no focal airspace opacity pleural effusion or pneumothorax the osseous structures are intact	the cardiomediastinal silhouette is normal in size and contour no focal consolidation pneumothorax or large pleural effusion negative for acute bone abnormality

Table 7. Generated reports for three test cases, showing the hierarchical LSTM, vanilla 6-layer Transformer (encoder unfrozen), and clinical-guided Transformer (D_6L, soft guidance with $\alpha_{\text{finding}} = 2.0$, $\beta = 0.5$). CXR1004 illustrates multi-view inconsistency: the clinical Transformer collapses to a generic template on view 0 but produces a longer report mentioning a real finding (*degenerative changes in the thoracic spine*) on view 1 of the same patient. CXR1015 illustrates the negation enumeration behavior produced by `decoding_bias` with `bias_include_negations=true`: the model emits an exhaustive list of common findings preceded by *no*, even when the reference describes positive findings (*bibasilar opacities*). The reference text is truncated for space; full references appear in the supplementary material. <UNK> denotes vocabulary tokens replaced during preprocessing.

Cell (All clinical transformer)	**LW**	**DB**	α_f	β	**B-1**	**B-4**	**MET**	**R-L**	**CIDEr**	**ChX$_{\text{macro}}$**
clinical_none_6layer	off	off	—	—	**0.301**	**0.096**	0.137	0.278	0.408	0.213
clinical_soft_finding_only_6layer	on	off	2.0	—	0.295	0.095	0.140	**0.283**	**0.507**	0.229
clinical_soft_bias_only_6layer	off	on	—	0.5	0.271	0.084	0.130	0.269	0.390	0.201
clinical_soft_6layer	on	on	2.0	0.5	0.269	0.084	0.135	0.279	**0.507**	0.233
clinical_hardguidance_6layer	on	on	3.0	1.0	0.279	0.073	**0.141**	0.272	0.374	**0.245**

Table 8. 6-layer ablation. `soft_6Layer` is the headline configuration reported in Section 4.3. Notation as in Table 2. Best per-column value bolded.